NetPlan calculation methodology

As of 14/08/2026

This document describes in full how the tool gets from imported data to a recommendation. Every formula corresponds to the code in `js/sim.js` (calculation core) and `js/dashboard.js` (aggregation across the network).

Die deutsche Fassung dieses Dokuments: `berechnungslogik.md`.

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1. The calculation path at a glance

```
Imported rows
   │  quantity logic (pallets per row)
   ▼
Demand per category and customer region  ──►  demand per day  ──►  target stock
   │                                                               │
   │  for each DC / region pair:                                   │ occupies
positions
   │  distance → transport cost, transit time                      │
   ▼                                                               ▼
Sub-score transport        Sub-score target coverage       Sub-score capacity
   └───────────────────────────────┬───────────────────────────────┘
                  weighted sum → total score per DC
                                   │
                  ranking → recommendation (single or split)
```

The calculation always runs for **one product category** and for the **data basis** and **period range** selected in the simulation. Allocations already applied for other categories occupy capacity and therefore influence the result for the next category.

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2. Step 1: From a data row to a pallet

Capacity is measured in pallet positions, so the tool converts every data row into pallets. The **first available** figure wins:

```
Pallets = pallet equivalent                    if > 0
        → else pallets                         if > 0
        → else volume ÷ volume per pallet      if > 0
        → else quantity ÷ units per pallet
```

Why this order: The pallet equivalent is the most accurate figure in business terms, because it already reflects packing density and stackability. Deriving from quantity is the crudest, because it assumes a uniform unit size across the whole category.

Both conversion factors live under *Data & Import → Quantity logic* and apply globally. Where categories differ strongly in packing density, pallets should be supplied in the data rather than derived from quantity.

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3. Step 2: Period length and demand per day

The analysis period is derived from the **periods actually contained** in the data, not from the number of rows:

```
Period length in days = sum of the days of all distinct periods in the filter
```

Each period counts with its real length: a month with 28 to 31 days, a quarter with 91.3, a calendar week with 7, a year with 365. Where a period cannot be resolved, 30.44 days are assumed (an average month).

```
Demand per day = pallets in the period ÷ period length in days
```

Important: Demand per day is an **average over the selected period**. Seasonal peaks are smoothed out by it. To size for the peak, restrict the period range to the peak months — demand per day then rises, and with it the target stock.

The value can also be set explicitly under *Analysis period (days)*, for example when the data has gaps.

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4. Step 3: Target stock

Target stock is the **capacity-relevant figure** — it occupies pallet positions:

```
Target stock = demand per day × target coverage × stock safety factor
```

- **Target coverage** in days, set per category or globally
- **Stock safety factor** as an uplift (1.0 = none, 1.1 = 10 % buffer)

Target coverage acts **linearly**: doubling it from 21 to 42 days doubles the positions required. It is therefore the single strongest lever in the whole model — stronger than any weighting.

Example (demo data, category Refrigeration, forecast over 12 months):

FIGURE	VALUE
Pallets in the period	140,016
Period length	365 days
Demand per day	383.6 pallets
Target coverage	28 days
Target stock	10,741 positions

5. Step 4: Distance, transit time, cost

Distance

```
Distance = great-circle distance (Haversine) × 1.28
```

The factor 1.28 is a flat detour factor approximating road distance against the great-circle line. Where a region has no coordinates, the **average known distance in the network** is substituted and the row is marked “estimated” in the interface.

Transport cost

```
Cost per pallet = base cost + cost per km × distance
```

A **region-specific flat rate stored on the DC overrides this calculation entirely** for that region. This is the recommended route where real freight rates are available: the distance-based formula is only a stand-in as long as no tariffs are on file.

```
Total transport cost = Σ (pallets of the region × cost per pallet)
```

Transit time

```
Transit time = fixed handling time + distance ÷ transport performance per day
```

Defaults: 0.5 days handling and 500 km/day.

Storage, handling and fixed cost

```
Storage cost  = target stock × cost per position and month × months in the period
Handling cost = pallets in the period × cost per pallet
Fixed cost    = fixed cost per period × number of periods
```

Storage cost is a **stock figure** (it follows target stock), handling cost a **throughput figure** (it follows volume). This is why a longer target coverage affects storage cost only, never handling.

Fixed costs feed into the network key figures on the dashboard, **not** into the site score of an individual category — otherwise the first category allocated would carry a site's entire fixed cost.

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6. Step 5: The three sub-scores

Each sub-score runs from 0 to 100. All three are shown individually in the ranking table and in the “Score composition” chart.

6.1 Capacity and balance

What matters is utilisation **after** the allocation:

```
Utilisation after = (already occupied + target stock) ÷ capacity
```

“Already occupied” covers the DC's base occupancy plus every allocation already applied for other categories.

```
Utilisation ≤ limit:      Score = 100 − 50 × (utilisation ÷ limit)
limit < utilisation ≤ 1:  Score = 50 × (1 − (utilisation − limit) ÷ (1 − limit))
Utilisation > 1:          Score = 0
```

With the default limit of 85 %:

UTILISATION AFTER	SCORE
0 % (empty)	100
42.5 %	75
85 % (limit)	50
92.5 %	25
100 %	0
above	0

The curve is continuous and monotonically decreasing. It rewards **headroom** and therefore acts as a balancing force: full sites become unattractive before they overflow. The utilisation limit is the point beyond which the penalty doubles in pace.

6.2 Transport and distance

```
Score = 100 × lowest cost per pallet in the field ÷ own cost per pallet
```

A **ratio scale**: the cheapest site always scores 100, twice the cost scores 50, three times the cost 33. Two consequences follow:

- The score is **relative to the field of candidates**. Remove a very cheap DC from the network and every remaining score rises.
- It is **scale-invariant**: multiply all cost rates uniformly and the ranking does not change.

6.3 Target coverage and stock

```
Stock capability = free positions ÷ target stock       (capped at 1)
Responsiveness   = 1 − transit time ÷ target coverage  (clamped to 0…1)

Score = 100 × (0.7 × stock capability + 0.3 × responsiveness)
```

The 70/30 split is fixed in the calculation core and not adjustable through the interface. The reasoning: whether the target stock fits into the warehouse at all matters more than a day more or less of transit time.

Responsiveness relates transit time to target coverage. At 35 days of coverage, 2 days of transit barely register (0.94); at 5 days of coverage the same transit weighs heavily (0.60).

---

7. Step 6: Weighting and total score

The three slider values are first normalised to sum to 1:

```
w_i = slider_i ÷ (slider_capacity + slider_transport + slider_coverage)
```

Only the **ratio** of the sliders matters, therefore: 30/45/25 gives exactly the same result as 60/90/50. If all three are set to 0, the tool falls back to one third each.

```
Total score = w_capacity × score_capacity
            + w_transport × score_transport
            + w_coverage × score_coverage
```

The ranking table shows the total score per site; the bar chart shows the three **contributions** (w_i × score_i), which add up to the total. It is therefore always visible which criterion carried the decision.

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8. What the weighting actually does – measured

The table below comes from a run over the bundled demo data set (category Refrigeration, data basis forecast, single allocation, empty network). “Gap” is the lead of the first-placed site over the second.

WEIGHTS (CAP./TRANSP./COV.)	RECOMMENDATION	SCORE	GAP	AVG. DISTANCE
100 / 0 / 0	**DC Poznań**	68.9	0.4	868 km
0 / 100 / 0	DC Duisburg	100.0	13.5	628 km
0 / 0 / 100	DC Duisburg	98.1	0.1	628 km
30 / 45 / 25 (default)	DC Duisburg	90.1	6.1	628 km
60 / 20 / 20	DC Duisburg	80.7	2.5	628 km
20 / 60 / 20	DC Duisburg	93.3	8.1	628 km
20 / 20 / 60	DC Duisburg	92.6	2.9	628 km
50 / 0 / 50	DC Duisburg	83.3	0.0	628 km

Four patterns emerge that hold generally:

First: the weighting decides less often than you would expect. In eight of ten combinations tested the same site wins. In a transport-cost-dominated network — and most are — proximity to customers overrides the other criteria almost every time. The recommendation only changes at 100 % capacity weight, and then by 0.4 points, which is effectively a tie.

Second: the gap says more than the score. A result with an 8.1-point lead is robust; one with a 0.0 to 0.4-point lead is a coin toss where other criteria should decide. Always check the row below the winner in the ranking.

Third: capacity weight lowers the level, not the order. Because the capacity score already sits well below 100 at moderate utilisation, the total score drops as soon as you raise that criterion (90.1 → 80.7 at 60 % weight). That is not a worse result, it is a different scale.

Fourth: the coverage score separates weakly while capacity suffices. At 0/0/100 the top three sites lie within 0.1 points of each other, because stock capability is 1.0 for all of them and only transit time differentiates. This criterion only develops its effect once capacity becomes scarce — then stock capability falls below 1 and the score drops quickly.

Which weighting to use when

SITUATION	RECOMMENDED EMPHASIS
Freight cost dominates, warehouses are well utilised	Transport high (50–70 %)
One site is overflowing, the network needs balancing	Capacity high (50–70 %)
Short delivery times promised, stock is tight	Coverage high (40–60 %)
First orientation without a prior commitment	Default 30 / 45 / 25

A sound approach is to run the same configuration with **two or three weightings** and save each as a scenario. If the allocation stays put, it is robust. If it flips, the scenario comparison shows what the alternative costs — and the decision gets made deliberately instead of by a slider.

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9. Single allocation and splitting

Single allocation

Every active DC is evaluated against the **full** demand of the category. The ranking follows the total score; the first place is the recommendation.

Splitting

Splitting works **region by region and greedily**:

1. First, all DCs are evaluated as in the single allocation. The best *n* form the candidate pool (*n* = “Max. number of DCs in the split”).
2. Customer regions are sorted by volume, largest first.
3. For each region the pool is evaluated again — using the **region-specific** transport cost and the capacity still **free** at that moment.
4. The region goes to the highest-scoring DC with free capacity. If that is not enough for the whole region, the fitting share is allocated and the remainder passes to the next-best DC in the following round.
5. If volume remains at the end because no capacity is free anywhere in the pool, it goes to the highest-scoring DC and is reported as overflow.

The procedure is **heuristic, not optimal**. It produces traceable, geographically sensible splits and respects capacity limits, but it does not guarantee a cost minimum in the sense of mathematical optimisation. For a deliberate deviation the shares can be overridden in the table and recalculated; each DC then serves the given percentage of **every** region.

Order in “Simulate all categories”

Categories are processed by volume, largest first. The highest-volume category therefore chooses first and finds an empty network; later categories compete for what remains. This order is deliberate, because misplacing the largest category would be the most expensive mistake. It also means: **the result depends on the order.** To set different priorities, allocate the categories individually in the order you want.

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10. History, forecast and both combined

Both data sets use the same schema and are processed identically. The difference lies **solely in which periods feed into demand per day**. The tool produces no forecast of its own and does not extrapolate.

Measured on the demo data set (category Refrigeration, default weighting):

DATA BASIS	PERIODS	LENGTH	PALLETS	DEMAND/DAY	TARGET STOCK	RECOMMENDATION
History	24	730 days	276,190	378.3	10,594	DC Duisburg
Forecast	12	365 days	140,016	383.6	10,741	DC Duisburg
History + forecast	35	1,064 days	416,205	391.2	10,953	DC Duisburg

What the choice does

History only reflects what actually happened: real customer structure, real regional distribution, real seasonality. It is the most reliable basis for the **regional allocation**, because it rests on measured shipments. Its weakness: it knows nothing about growth, range shifts, or customers won and lost. Sizing a network for the coming years on history alone tends to size it too small.

Forecast only reflects the planned future and is therefore the right choice for an investment or site decision. The quality of the result is, however, exactly the quality of the forecast. **Regional granularity** is the critical point: many forecasts exist only at country or total level. Without the regional split the distance calculation loses its basis and all sites converge in the transport score.

Both combined averages across the whole period. Demand per day then lands between the two values, weighted by the number of days — in the example the history, being twice as long, pulls the figure down. That is useful for damping outliers in the forecast, and misleading where history and forecast are structurally different: an average of two different worlds describes neither.

Recommended approach

1. Run **history** to understand the regional structure and the order of magnitude, and to calibrate the cost rates.

2. Run **forecast**, because the decision concerns the future — save it as a scenario.
3. Compare both scenarios. Where the allocations differ lies the real insight: the network is not robust against how demand develops. A split variant or a capacity adjustment is then worth examining.

The “both” combination is mainly useful when the forecast spans only a few periods and would otherwise be dominated by a single season.

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11. Limits: time horizon and data volume

The figures below were measured on the standard Chromium of the build environment.

Time horizon

There is no limit on the number of forecast periods. A data set of 10 years of history (2016-01 to 2025-12) and 10 years of forecast (2026-01 to 2035-12) was tested — 240 monthly periods, processed without error, simulated across the full 3,652 forecast days.

Limits exist only in **period recognition**:

NOTATION	VALID RANGE
`2035-06`, `2035-06-15`	practically unlimited (tested to year 2500)
`15.06.2035`, `06/2035`	as above
`2035-Q3`, `2035-KW26`	as above
`Jun 2035`	as above
bare year: `2035`	**1900 to 2200**
Excel date value	approx. 1954 to 2064 (serial numbers 20,000–60,000)

The practical limit is a business one: a forecast beyond three to five years rarely carries a site decision, and the longer the selected period, the more the average smooths out seasonality.

Historical data volume

Here too there is **no fixed row limit**. Measured values:

RECORDS	SIMULATING ONE CATEGORY	ALL CATEGORIES	PROJECT SIZE	BROWSER CACHE
5,000	2 ms	1.2 s	1.4 MB	works
25,000	6 ms	0.8 s	6.7 MB	**limit exceeded**
100,000	44 ms	1.3 s	26.7 MB	limit exceeded
250,000	109 ms	2.6 s	66.7 MB	limit exceeded

Computation time is not the constraint: even 250,000 records are evaluated in fractions of a second. The real limit is the browser's **localStorage at about 5 MB**, roughly **15,000 to 18,000 records**. Beyond that the built-in fallback takes over: configuration, allocations and scenarios continue to be saved, raw data is not, and a corresponding notice appears. Nothing is lost — the working state simply has to be secured via *Export & Project → Save project as JSON*.

Two things ease the pressure:

- On import, files with more than 2,000 rows are automatically condensed to the dimension combination (period × customer × region × category). A million individual deliveries often collapse into a few tens of thousands of records.
- The simulation works on the aggregated level anyway. Condensing before import costs no accuracy as long as the dimensions are preserved.

Recommendation: if the file yields more than roughly 20,000 condensed records, switch off automatic caching and work with the project file instead.

Format for the forecast upload

The forecast uses **exactly the same format as the history** — there is no separate forecast schema. The only distinction is the *Data set type: Forecast* switch in the import area.

Required per row:

FIELD	EXAMPLE
Product category	Refrigeration
Period	2027-03
one quantity measure	quantity, pallets, pallet equivalent, volume or revenue
one location reference	region, country or customer

Optional: customer, country, quantity, revenue, volume, pallets, pallet equivalent, latitude and longitude.

File types: `.xlsx`, `.xlsm`, `.xls`, `.csv`, `.txt`. Column names are free — they are mapped during import and common labels are detected automatically. Ready-to-fill templates sit in `vorlagen/`; the Excel workbook contains a dedicated “Forecast” sheet for this.

Two notes from practice:

- **Supply the regional split.** A forecast at country level works, but makes the distance evaluation coarse. The finer the regions, the more reliable the site recommendation.
- **Forecast in pallets or pallet equivalents where possible.** If planning is done in units or revenue only, the entire capacity requirement hangs on a single global conversion factor.

12. What the model does not do

So that the results are placed correctly:

- **No mathematical optimisation.** Splitting is a traceable heuristic, not a solver. It finds good solutions, not provably optimal ones.
- **No route planning.** Distances are great-circle times a detour factor. For reliable freight costs the region-specific flat rates per DC are the intended route.
- **No forecasting.** The forecast is produced externally and only read in.
- **No inventory optimisation.** Target stock follows the coverage you specify; safety stock derived from service level and forecast error is not calculated. The stock safety factor is a flat stand-in for that.
- **No time dynamics.** The calculation is an average across the selected period, not a profile over the periods. Seasonal capacity peaks only become visible if you restrict the period accordingly.
- **No multi-echelon view.** What is modelled is the customer region ← DC leg. Inbound cost from plants or central warehouses into the DCs is not part of the model; it can be approximated through the handling cost per pallet.